COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit 3 : Session 11 : SLO 1 — IEEE 754 Single Precision Format
Name: JATANPREET SINGH VIRDI JATANPREET SINGH VIRDI | Reg. No: RA2511003011374 | Solved Worksheet

Section A: Match the Following (Remember)

• A. Sign Bit → 2. Indicates positive or negative number
• B. Exponent Bias → 3. 127 in single precision
• C. Fraction (Mantissa) → 1. 23 bits used for precision representation
• D. Total bits in IEEE 754 (Single) → 5. 32 bits total
• E. Exponent Field → 4. Converts decimal exponent to binary

Format Summary

• IEEE 754 single precision = 32 bits: 1 sign bit (bit 31), 8 exponent bits with bias 127 (bits 30-23), 23 mantissa/fraction bits (bits 22-0).
• Value = (-1)^sign × 1.mantissa × 2^(exponent - 127) for normalized numbers.